

Radar-Assisted Compressed Sensing for Enhanced Channel Estimation in V2X Communications

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Abstract—Channel estimation is a fundamental component in modern vehicular-to-everything (V2X) communication systems, supporting efficient and reliable data transmission in dynamic and often resource-constrained environments. Traditional methods of channel estimation are resource-intensive, relying on significant amounts of training data and computational power, which limits their applicability in real-time V2X scenarios. Compressed Sensing (CS) has emerged as a promising solution to alleviate these demands by reducing both training data requirements and computational complexity. However, in dense urban environments characterized by prevalent non-line-of-sight (NLOS) conditions, traditional CS algorithms are challenged by complex multipath effects and signal degradation. To overcome these challenges, we propose a novel radar-assisted CS framework that leverages radar sensing to enhance the adaptability and accuracy of channel estimation under strong NLOS conditions. Our simulation results validate the efficacy of this approach, demonstrating marked improvements in both estimation accuracy and reliability across challenging V2X communication scenarios.

Index Terms—V2X Communications, Channel Estimation, Compressed Sensing, Integrated Sensing and Communication

I. INTRODUCTION

Rapid advancement of vehicle-to-everything (V2X) applications has become a central focus in 5G and beyond, driving significant research interest due to the need for reliable, high-speed, and low-latency communication between vehicles and their surroundings [1]. A critical requirement in these urban and high-mobility environments is accurate channel estimation, which supports effective data transmission and overall network performance. However, non-line-of-sight (NLOS) propagation and rapid channel variations in urban landscapes present considerable challenges for conventional channel estimation techniques [2]–[4], as these methods are often hampered by obstructions, reflections, and computational constraints. Integrated sensing and communication (ISAC) [5] systems offer a promising dual function approach utilizing shared waveforms for both radar sensing and communication purposes. ISAC systems enable cooperative radar and communication functionalities, where radar can identify obstacles such as buildings, derive environmental dynamics, and assist in overcoming challenges associated with NLOS conditions. This integration provides valuable information on the surrounding environment, enabling improved channel estimation performance even under adverse conditions.

In multipath-dense urban settings, the inherent sparsity of wireless channels makes compressed sensing (CS) an appealing solution for channel estimation. CS techniques re-

duce training data requirements by focusing on the most significant signal paths, making them suitable for resource-constrained and high-mobility scenarios [6]. Leveraging CS within an ISAC framework, previous studies have proposed two prominent radar-assisted channel estimation methods. The first approach employs a two-stage method using the simultaneous weighting orthogonal matching pursuit (SWOMP) algorithm with sparse Bayesian learning (SBL) [7], [8]. Initially, SWOMP identifies active paths and angles of arrival (AoA) [9] based on radar-derived sensing information. The subsequent SBL stage refines delay estimates and calculates channel gains. Although effective, this method incurs high computational complexity, particularly in high-dimensional, large-scale MIMO systems, which may limit its real-time applicability. The second approach addresses the orthogonal time frequency space (OTFS) [10] channel estimation problem as a sparse recovery task, where radar data is used to establish a support set, identifying locations of non-zero delay-Doppler taps. The orthogonal matching pursuit (OMP) algorithm utilizes propagation delays, Doppler frequencies [11], and AoA derived from radar to improve sparse OTFS channel recovery. However, under high-noise conditions, the OMP algorithm can mistakenly select incorrect basis vectors, leading to cumulative estimation errors, especially in dynamic V2X environments.

To overcome these limitations, we introduce a novel and computationally efficient approach dubbed sensing-assisted simultaneous weighting orthogonal matching pursuit (SA-SWOMP). Our algorithm directly integrates radar sensing data for support set determination, optimizing the SWOMP process by accurately identifying non-zero delays and angles of arrival. Using radar-derived data for initial support set determination, the SA-SWOMP algorithm reduces search space and minimizes computational demands, particularly beneficial in MIMO-OFDM systems [12]. Moreover, by integrating radar-based support with the SWOMP weighting mechanism, our method improves resistance to noise, enhancing both accuracy and environmental adaptability in time-varying scenarios, such as those encountered in vehicular networks.

In summary, our main contributions in this paper are as follows:

- 1) To the best of our knowledge, this work is the first to design a CS algorithm with a radar-informed initial support set, enabling channel estimation to begin with a more accurate starting point. This approach effectively reduces search complexity and improves estimation accuracy by

focusing on significant paths from the outset.

- 2) By leveraging SWOMP's weighted measurement approach, we enhance it with radar-based scatter sensing, which mitigates the risk of selecting incorrect basis vectors under noisy conditions. This improvement enhances the algorithm's robustness and reduces the likelihood of cumulative errors.
- 3) Comprehensive simulations demonstrate the superior performance of our proposed radar-assisted CS framework in terms of accuracy, robustness, and computational efficiency when compared to existing methods, validating its applicability in urban V2X communication systems.

The remainder of this paper is organized as follows. In Section II, we describe the system model used to characterize the urban propagation environment. Section III details the design of our proposed sensing-assisted channel estimation algorithm. The simulation results and analysis are provided in Section IV. Finally, conclusions are drawn in Section V.

II. SYSTEM MODELS AND SENSING INFORMATION

In this section, we first present the MIMO-OFDM [13] system model used in this study. We then describe the radar sensing information utilized to enhance channel estimation.

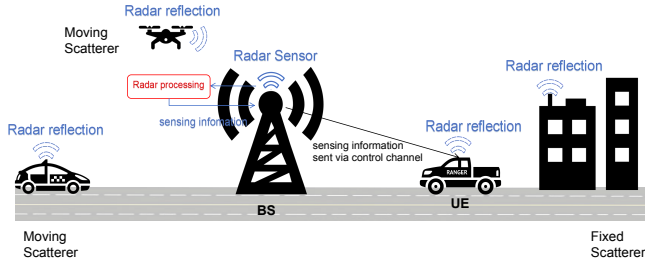


Fig. 1. System model illustrating how the BS leverages radar sensing data for channel estimation.

As shown in Fig. 1, we consider a communication system in which a base station (BS) serves a single, highly mobile user equipment (UE) at a carrier frequency f_c . The BS is equipped with a uniform linear array (ULA) [14] of antennas for communication with the UE, and also includes a frequency-modulated continuous-wave (FMCW) radar [15] operating at a start frequency f_0 . The radar collects environmental sensing information, which is then used to support downlink channel estimation [16]. For simplicity, we assume that the UE has a single antenna. However, the proposed sensing-assisted MIMO-OFDM channel estimation method can be extended to systems with multi-antenna UEs.

B. Sensing Information

We consider a scenario where a UE communicates with a BS in an environment populated by scatterers. Since the

environment is dynamic, the positions of these scatterers vary and are represented as:

$$\mathbf{p} = \{p_1, p_2, \dots, p_L\}, \quad (1)$$

where $p_i = (x_i, y_i)$ represents the position of the i -th scatterer, and L is the total number of scatterers. We assume that the positions of the scatterers are random and follow a uniform distribution within a defined two-dimensional space.

The positions of the scatterers can be generated as:

$$\begin{aligned} x_i &\sim \mathcal{U}(-r, r), \\ y_i &\sim \mathcal{U}(-r, r), \end{aligned} \quad (2)$$

where $\mathcal{U}(-r, r)$ denotes a uniform distribution over the interval $[-r, r]$, with r representing the radar detection range.

The BS obtains sensing information either via a co-located radar operating in a separate band [17] or through a joint communication-sensing framework [18]. The sensing information obtained includes the time delay τ and the angle of arrival θ .

Given a scatterer position $p_\ell = (x_\ell, y_\ell)$, BS position $p_{BS} = (x_{BS}, y_{BS})$, and UE position $p_{UE} = (x_{UE}, y_{UE})$, the time delay and angle are calculated as follows:

1) *Time Delay τ* : The time delay τ_ℓ represents the round-trip time for the radar signal to travel from the BS to the ℓ -th scatterer and back. Assuming the speed of light is c , the distance from the BS to the scatterer is:

$$d_\ell = \sqrt{(x_\ell - x_{BS})^2 + (y_\ell - y_{BS})^2}, \quad (3)$$

and the corresponding round-trip time delay is:

$$\tau_\ell = \frac{2d_\ell}{c} = \frac{2\sqrt{(x_\ell - x_{BS})^2 + (y_\ell - y_{BS})^2}}{c}. \quad (4)$$

2) *Angle of Arrival θ* : The AoA of the ℓ -th scatterer relative to the BS is calculated as:

$$\theta_\ell = \arctan\left(\frac{y_\ell - y_{BS}}{x_\ell - x_{BS}}\right). \quad (5)$$

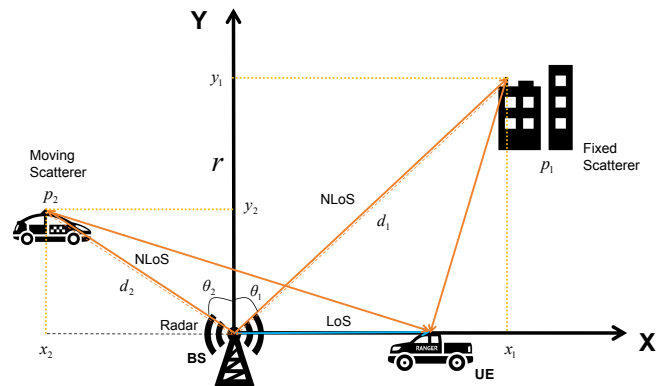


Fig. 2. Propagation paths for radar sensing and downlink communication signals in the system model.

As illustrated in Fig. 2, the radar detection range is r . With the known positions of the BS, UE, and scatterers

$(p_1, p_2, \dots, p_L)$, the radar can obtain AoA and delay information. Incorporating this time-varying data into the channel estimation process allows for improved robustness and accuracy in challenging communication environments.

III. PROPOSED CHANNEL ESTIMATION ALGORITHM

In vehicular communication environments, channel characteristics exhibit significant sparsity, where most channel coefficients are zero or near-zero, and only a few coefficients hold substantial values. Traditional channel estimation methods struggle with such sparsity, often resulting in poor performance. Compressed sensing methods, however, can efficiently exploit this sparsity for improved estimation. The SWOMP algorithm is a CS-based method particularly suited for sparse channels. A distinguishing feature of SWOMP is its autonomous weight selection capability, where the weight matrix $\mathbf{W}$ plays a critical role. This matrix adjusts and optimizes the algorithm, allowing it to adapt dynamically to various channel conditions, which significantly enhances estimation accuracy and robustness. Such adaptability makes the SWOMP algorithm highly effective in the complex and time-varying environments typical of vehicular communication. In this section, we present a sparse channel recovery method that leverages radar-derived sensing information to improve the recovery of sparse channels. By incorporating sensing information, our approach further enhances the performance of the SWOMP algorithm in challenging vehicular communication environments. This integration enables the SWOMP algorithm to more accurately and efficiently estimate sparse channels, ultimately leading to more robust and reliable communication.

A. Sensing Information Formulation

By processing radar data, we can obtain a sparse channel representation that includes key parameters such as path gain, delay, and angle of arrival. The radar processing steps [19] are as follows:

- **Filtering and Denoising:** Noise is removed from the radar signals to enhance the accuracy of subsequent processing steps.
- **Pulse Compression:** Matched filtering is applied to compress radar pulses, improving range resolution.
- **Range-Doppler Processing:** Fast Fourier Transform (FFT) is used to convert radar signals into the range-Doppler domain, providing distance information for targets.
- **Target Detection:** Constant false alarm rate (CFAR) detection [20] is implemented to identify significant targets in the radar data.
- **Parameter Estimation:** For each detected target, the path gain, delay, and AoA are estimated.

The output of radar processing is a set of parameters that describe the sparse channel, which is then utilized in the channel estimation algorithm.

The radar sensing information, denoted by $\mathbf{s}$, represents a set of indices corresponding to potential non-zero elements in the channel matrix. These indices are derived from environmental

information obtained by the radar (e.g., delay and angle). Mathematically, the radar data can be mapped to specific locations in the channel matrix, providing valuable insights for identifying the sparse structure of the channel.

1) *Conversion of Delay and Angle:* Assume the radar captures the following information about scatterers in the environment:

- **Delay τ_ℓ :** The propagation delay from the ℓ -th scatterer.
- **Angle θ_ℓ :** The angle of arrival or departure for the ℓ -th scatterer.

These parameters can be mapped to positions in the channel matrix by converting the physical parameters into corresponding matrix indices.

Delay Index:

The delay τ_ℓ is converted to the delay tap index m_ℓ using the following formula:

$$m_\ell = \text{round}(N_d \Delta f \tau_\ell), \quad (6)$$

where N_d represents the number of subcarriers, Δf is the subcarrier spacing, and τ_ℓ is the delay obtained from radar sensing.

Angle Index:

The angle θ_ℓ is converted to the index associated with the angular component of the channel as:

$$f_\ell = \arg \max_f |\mathbf{a}(\theta_\ell)|, \quad (7)$$

where $\mathbf{a}(\theta_\ell)$ denotes the antenna array response vector, which corresponds to the array's directional pattern.

2) *Representation of Radar-Derived Support Set $\mathbf{s}$:* By combining the indices m_ℓ and f_ℓ , $\mathbf{s}$ can be represented as a set of indices corresponding to potential non-zero elements in the channel matrix. For each scatterer ℓ , the associated index t_ℓ is defined as:

$$t_\ell = f_\ell + (A - 1)m_\ell, \quad (8)$$

where A represents the number of base station antennas.

The radar-derived support set $\mathbf{s}$ is then expressed as a set of indices:

$$\mathbf{s} = \{t_1, t_2, \dots, t_L\}, \quad (9)$$

where L denotes the number of scatterers detected by the radar, and each t_ℓ corresponds to a potential non-zero location in the channel matrix, inferred from radar sensing information.

In this way, $\mathbf{s}$ serves as prior knowledge to guide the column selection process in the SWOMP algorithm, enhancing its efficiency and accuracy in identifying significant channel components.

B. Sensing-Assisted SWOMP Algorithm (SA-SWOMP)

Fig. 3 illustrates the signal processing flow in a system utilizing our proposed SA-SWOMP algorithm for estimating the channel coefficient. This approach processes the transmitted signal and radar data simultaneously, allowing for efficient and

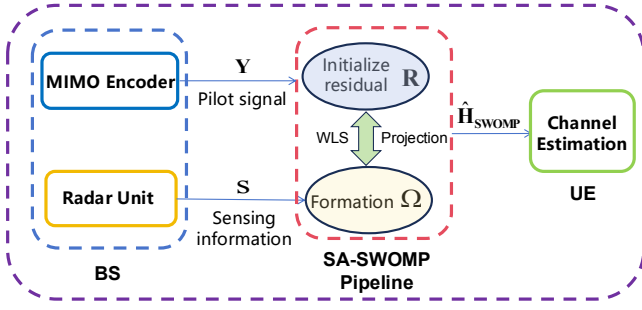


Fig. 3. Signal processing flow diagram of the SA-SWOMP algorithm for channel estimation.

accurate channel state information estimation while reducing the complexity of the SWOMP algorithm.

We propose using the radar-derived sensing information s to enhance sparse channel recovery. Algorithm 1 presents the detailed structure of SA-SWOMP, an improved version of the SWOMP algorithm tailored for sparse channel recovery. The sensing information s initializes the estimated support set of the sparse signal. The delay and angle-of-arrival information from radar sensing is assumed to closely match the communication channel characteristics.

Algorithm 1 SA-SWOMP Algorithm

- 1: **Input:** Received pilot signal $\mathbf{Y}$, DFT matrix Ψ , measurement matrix Φ , maximum iterations T , weight matrix $\mathbf{W}$, sensing information s .
 - 2: **Output:** Estimated channel matrix $\hat{\mathbf{H}}_{\text{SWOMP}}$.
 - 3: Compute the equivalent observation matrix $\mathbf{A} = \Psi\Phi$.
 - 4: Initialize residual $\mathbf{R} = \mathbf{Y}$, support set $\Omega = s$,
 - 5: **for** $t = 1$ to T **do**
 - 6: Find column indices $\mathbf{i}^* = [i_0^*, i_1^*, \dots, i_{k-1}^*]$ that maximize the correlation: $\mathbf{i}^* = \arg \max_{i \in s} |\langle \mathbf{A}_i, \mathbf{R} \rangle|$, where k is the number of indices, each unique.
 - 7: Update support set: $\Omega = \Omega \cup \{\mathbf{i}^*\}$.
 - 8: Project the non-zero coefficients of the signal onto the subspace given by the support using weighted least squares (WLS):
 - 9: $\hat{\mathbf{H}}_{\Omega} = \arg \min_{\mathbf{H}} \|\mathbf{W}(\mathbf{Y} - \mathbf{A}_{\Omega}\mathbf{H})\|^2$.
 - 10: Update residual: $\mathbf{R} = \mathbf{Y} - \mathbf{A}_{\Omega}\hat{\mathbf{H}}_{\Omega}$.
 - 11: If stopping criterion is met, break.
 - 12: **end for**
 - 13: Reconstruct the full channel estimate $\hat{\mathbf{H}}_{\text{SWOMP}}$ using $\hat{\mathbf{H}}_{\Omega}$ and support set Ω .
-

SA-SWOMP iteratively updates the estimated support of the sparse signal by identifying the strongest correlation between the residual signal and the sensing matrix. In each iteration, a weighted least squares (WLS) estimate [4] is applied to the signal elements indicated by the estimated support. Here, the weight matrix $\mathbf{W}$ is defined as the inverse of the square root of the noise covariance matrix, $\mathbf{C}_0$. This ensures that residuals are weighted according to their noise variance, giving greater

weight to components with lower noise. Due to multipath effects causing signal overlap in MIMO systems, the noise covariance matrix may be full rank. Thus, we use the Cholesky decomposition of $\mathbf{C}_0$:

$$\mathbf{W}^T \mathbf{W} = \mathbf{C}_0^{-1}, \quad (10)$$

where $\mathbf{W}$ can be obtained as an upper triangular matrix. This weighting approach allows the SA-SWOMP algorithm to prioritize measurements with lower noise, thereby enhancing its adaptability to different channel environments and improving overall performance.

The radar-derived support set Ω provides a reliable starting point. By initializing Ω with radar-derived indices, the algorithm reduces its reliance on random or blind search, quickly focusing on dominant paths, especially in environments with sparse scatterers. Restricting the search to a smaller set of columns also reduces the computational complexity in each iteration. Radar sensing ensures that the algorithm does not overlook critical channel components, leading to more accurate channel reconstruction and improved communication reliability.

IV. SIMULATION RESULTS

In this section, we present simulation results to evaluate the performance of our proposed solution. We compare various channel estimation methods, including Minimum Mean Squared Error (MMSE) [3], Orthogonal Matching Pursuit (OMP), Simultaneous Orthogonal Matching Pursuit (SOMP), and SWOMP, across different signal-to-noise ratios (SNRs).

For the simulations, we assume the system has 8 transmit antennas, denoted as $N_t = 8$, and a single receive antenna, denoted as $N_r = 1$. The key simulation parameters are summarized in Table I.

TABLE I
KEY SIMULATION PARAMETERS

Parameter	Value
Channel Type	Multipath Rayleigh Channel
Modulation Order	2
FFT Size (N_{fft})	256
Guard Interval Length (N_g)	32
Pilot Interval (N_{ps})	32
Number of Pilots (N_p)	8
Data Subcarriers (N_d)	248
Number of scatterers (L)	8

The simulations were conducted on a high-performance workstation with an Intel Core i9-12900H CPU and 16 GB of RAM. MATLAB version R2020b was used for all simulations and data analysis.

A. Comparative Analysis of Algorithms

In traditional MIMO-OFDM channels, we conducted a performance comparison of state-of-the-art algorithms. Specifically, we compared SWOMP with MMSE, OMP, and SOMP algorithms.

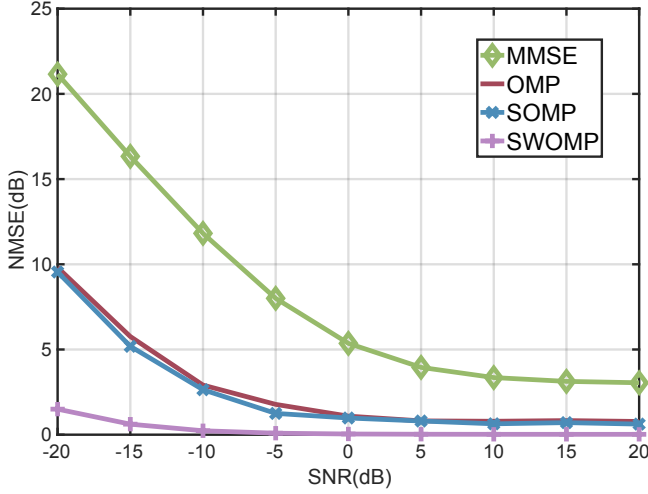


Fig. 4. Normalized Mean Squared Error (NMSE) comparison of MMSE, OMP, SOMP, and SWOMP algorithms.

In Fig. 4, we observe that at low SNR levels, noise introduces relatively large estimation errors. However, as the SNR increases, the impact of noise on the Normalized Mean Squared Error (NMSE) diminishes, leading to NMSE convergence.

In MIMO channels, the performance of the CS-based algorithms shows a notable improvement over traditional channel estimation methods. In urban environments, especially in vehicular communication scenarios, the channel is highly sparse. Consequently, channel estimation using CS algorithms significantly outperforms traditional methods, demonstrating strong robustness against noise in sparse channels. Notably, even when the SNR is below 0 dB, the CS algorithm achieves relatively low estimation error. Among the CS algorithms, SWOMP demonstrates superior noise resistance due to its utilization of weighted coefficients, which aligns with the results obtained in this study.

B. Comparative Analysis of Sensing Information-aided Method

Next, we examine the performance improvement of the three CS algorithms with the inclusion of radar sensing data.

In Fig. 5, the curves labeled “SA-OMP,” “SA-SOMP,” and “SA-SWOMP” represent the results of the OMP, SOMP, and SWOMP algorithms with sensing assistance, respectively. The error of each CS algorithm is further reduced after incorporating radar information. This enhancement aligns with the requirements for high precision, accuracy, and robustness in vehicular communications, improving the overall performance of the channel estimation algorithms. Notably, this improvement contributes to increased driving safety and efficiency. As the SNR increases, the error gradually decreases and stabilizes. However, the proposed SA-SWOMP algorithm, which leverages radar sensing information, continues to demonstrate advantages that were not observed in the previous algorithms.

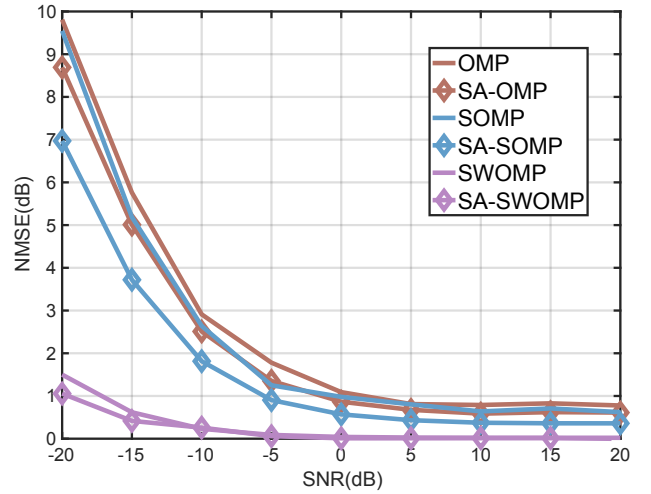


Fig. 5. Error comparison of SA-OMP, SA-SOMP, and SA-SWOMP algorithms with radar sensing data.

C. Comparative Analysis of Different Scatterer Quantities

In traditional algorithms, an increase in scatterer quantity typically reduces accuracy due to interference caused by the scatterers. However, by using scatterer information as prior knowledge, we incorporated it into the algorithm’s support set, transforming the scatterer information into a factor that enhances the algorithm’s accuracy and adaptability to the complexity of urban environments. As shown in Fig. 6, we compare the performance of our proposed SA-SWOMP algorithm with varying numbers of scatterers: 8, 16, 32, and 64.

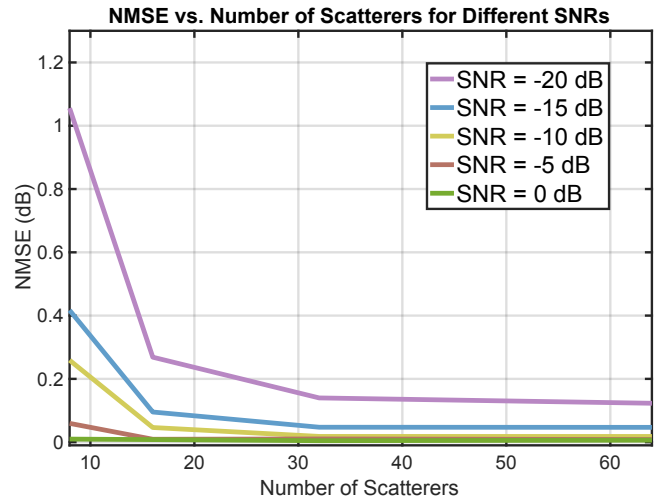


Fig. 6. Error comparison of the SA-SWOMP algorithm with varying numbers of scatterers.

We selected an SNR range from -20 dB to 0 dB for comparison, as the low SNR in this range makes the errors more pronounced. The plot shows that as the number of scatterers increases, the error decreases, indicating that

the SA-SWOMP algorithm effectively meets the complexity requirements of urban environments. This approach further enhances the robustness of wireless communication in urban settings, enabling the system to manage interference-heavy scenarios effectively, addressing the issue of excessive signal interference in dense urban areas.

D. Computational Complexity

The SWOMP algorithm has a computational complexity of $\mathcal{O}(T(KkN + T^2N + T^3))$, while the SA-SWOMP algorithm with radar sensing has a complexity of $\mathcal{O}(T(skN + T^2N + T^3))$. These complexities depend on several parameters: T , N , K , k , and s , where N and K are the dimensions of the measurement matrix, T is the number of iterations, k is the number of atoms selected per iteration, and s is the number of support sets that influence algorithm performance.

In the traditional SWOMP algorithm, all K columns of the sensing matrix are considered. However, radar sensing provides prior information about the sparsity of the signal, allowing us to identify columns likely to contain significant signal components and ignore a large number of irrelevant columns. Consequently, radar sensing effectively narrows the search range, reducing the number of columns s that need to be considered, which is significantly smaller than the total number of columns K . This approach substantially reduces the algorithm's complexity while maintaining high recovery quality.

V. CONCLUSION

In this paper, we introduced a novel channel estimation algorithm enhanced by radar sensing information to improve the accuracy of channel estimation. This algorithm combines environmental insights gathered from radar sensing with compressed sensing methods for channel estimation, enabling accurate assessment of channel states without relying on a large number of pilot signals. Key advantages of this approach include reduced pilot overhead in MIMO systems and improved estimation performance, both of which are crucial for optimizing spectral efficiency and minimizing system complexity. To evaluate the effectiveness of the proposed algorithm, we conducted extensive simulations comparing its performance with traditional channel estimation techniques across various conditions. The results consistently demonstrated that our sensing-aided channel estimation algorithm achieves higher accuracy, regardless of the complexity of the propagation environment. Integrating radar sensing into channel estimation thus represents a promising direction for enhancing the efficiency and performance of wireless communication systems.

Future research could explore adaptive sensing strategies that adjust radar parameters in real time, extend the approach to multi-user and multi-cell environments to manage interference, and implement hardware tests to evaluate practical performance and scalability in real-world applications.

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REFERENCES

- [1] J. Wang, Y. Shao, Y. Ge, and R. Yu, "A survey of vehicle to everything (v2x) testing," *Sensors*, vol. 19, no. 2, p. 334, 2019.
- [2] M. P. Wylie and J. Holtzman, "The non-line of sight problem in mobile location estimation," in *Proceedings of ICUPC-5th International Conference on Universal Personal Communications*, vol. 2. IEEE, 1996, pp. 827–831.
- [3] M. Tüchler, A. C. Singer, and R. Koetter, "Minimum mean squared error equalization using a priori information," *IEEE Transactions on Signal Processing*, vol. 50, no. 3, pp. 673–683, 2002.
- [4] Y. S. Cho, J. Kim, W. Y. Yang, and C. G. Kang, *MIMO-OFDM wireless communications with MATLAB*. John Wiley & Sons, 2010.
- [5] F. Liu, Y. Cui, C. Masouros, J. Xu, T. X. Han, Y. C. Eldar, and S. Buzzi, "Integrated sensing and communications: Toward dual-functional wireless networks for 6g and beyond," *IEEE journal on selected areas in communications*, vol. 40, no. 6, pp. 1728–1767, 2022.
- [6] D. L. Donoho, "Compressed sensing," *IEEE Transactions on information theory*, vol. 52, no. 4, pp. 1289–1306, 2006.
- [7] J. Rodríguez-Fernández, N. González-Prelcic, K. Venugopal, and R. W. Heath Jr, "Frequency-domain compressive channel estimation for frequency-selective hybrid mmwave mimo systems," *arXiv preprint arXiv:1704.08572*, 2017.
- [8] M. E. Tipping, "Sparse bayesian learning and the relevance vector machine," *Journal of machine learning research*, vol. 1, no. Jun, pp. 211–244, 2001.
- [9] L. Chen, I. Ahriz, and D. Le Ruyet, "Aoa-aware probabilistic indoor location fingerprinting using channel state information," *IEEE internet of things journal*, vol. 7, no. 11, pp. 10 868–10 883, 2020.
- [10] R. Hadani and A. Monk, "Otfis: A new generation of modulation addressing the challenges of 5g," *arXiv preprint arXiv:1802.02623*, 2018.
- [11] A. Amar and A. J. Weiss, "Localization of narrowband radio emitters based on doppler frequency shifts," *IEEE Transactions on Signal Processing*, vol. 56, no. 11, pp. 5500–5508, 2008.
- [12] H. Tang, J. Wang, and L. He, "Off-grid sparse bayesian learning-based channel estimation for mmwave massive mimo uplink," *IEEE Wireless Communications Letters*, vol. 8, no. 1, pp. 45–48, 2018.
- [13] Y. G. Li and G. L. Stuber, *Orthogonal frequency division multiplexing for wireless communications*. Springer Science & Business Media, 2006.
- [14] Y. Bresler and A. Macovski, "On the number of signals resolvable by a uniform linear array," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 34, no. 6, pp. 1361–1375, 1986.
- [15] A. G. Stove, "Linear fmcw radar techniques," in *IEE Proceedings F (Radar and Signal Processing)*, vol. 139, no. 5. IET, 1992, pp. 343–350.
- [16] Y. Gu and Y. D. Zhang, "Information-theoretic pilot design for downlink channel estimation in fdd massive mimo systems," *IEEE Transactions on Signal Processing*, vol. 67, no. 9, pp. 2334–2346, 2019.
- [17] H. Li and B. Himed, "Transmit subaperturing for mimo radars with co-located antennas," *IEEE Journal of Selected Topics in Signal Processing*, vol. 4, no. 1, pp. 55–65, 2010.
- [18] M. L. Rahman, J. A. Zhang, X. Huang, Y. J. Guo, and R. W. Heath, "Framework for a perceptive mobile network using joint communication and radar sensing," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 3, pp. 1926–1941, 2019.
- [19] F. Engels, P. Heidenreich, M. Wintermantel, L. Stäcker, M. Al Kadi, and A. M. Zoubir, "Automotive radar signal processing: Research directions and practical challenges," *IEEE Journal of Selected Topics in Signal Processing*, vol. 15, no. 4, pp. 865–878, 2021.
- [20] R. Nitzberg, "Constant-false-alarm-rate signal processors for several types of interference," *IEEE Transactions on Aerospace and Electronic Systems*, no. 1, pp. 27–34, 1972.